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Compare permissions on Regular Files and Directories

Create a regular file named sample.txt and a directory named sample_dir. Create at least one file inside sample_dir.

Apply each of the following permissions using chmod. Perform the specified operations and record your observations.

Note: Perform the exercise as a normal user. Do not use sudo.

Permission	Regular File (sample.txt) – Test	Observation	Directory (sample_dir) – Test	Observation
400 (r--)	Try to read and modify the file	Read succeeds; modification fails.	Try to list and enter the directory	Listing succeeds; entering succeeds if directory has r but not x? Correction: listing succeeds, entering fails because x is required.
200 (-w-)	Try to read and modify the file	Read fails; writing/modification succeeds (subject to how the program opens the file).	Try to list, enter, and create a file	Listing fails; entering fails; creating files fails because x is required to access directory entries.
100 (--x)	Try to read and execute the file	Reading fails; execution may succeed only if the file is a valid executable/script and the required interpreter/access conditions are met.	Try to list and enter the directory	Listing fails; entering succeeds.
500 (r-x)	Try to read and modify the file	Reading succeeds; modification fails.	Try to list and enter the directory	Listing succeeds; entering succeeds.
600 (rw-)	Try to read and modify the file	Reading and modification succeed.	Try to list, enter, and create a file	Listing succeeds; entering fails; creating a file fails.
700 (rwx)	Try to read, modify, and execute the file	Reading and modification succeed; execution is allowed by the permission bits (assuming the contents are executable).	Try to list, enter, create, and delete files	All operations succeed.

1. 400 (r--) Try to read and modify the file__Try to list and enter the directory

400 (r--) — Read Only

```
srinivas@DESKTOP-E4HFS1K:~$ ls
srinivas@DESKTOP-E4HFS1K:~$ touch sample.txt
srinivas@DESKTOP-E4HFS1K:~$ mkdir sample_dir
srinivas@DESKTOP-E4HFS1K:~$ touch sample_dir/inside.txt
srinivas@DESKTOP-E4HFS1K:~$ ls -l
total 0
-rw-rw-r-- 1 srinivas srinivas  0 Aug 22 12:17 sample.txt
drwxrwxr-x 1 srinivas srinivas 4096 Aug 22 12:18 sample_dir
srinivas@DESKTOP-E4HFS1K:~$ chmod 400 sample.txt
srinivas@DESKTOP-E4HFS1K:~$ chmod 400 sample_dir
srinivas@DESKTOP-E4HFS1K:~$ cat sample.txt
srinivas@DESKTOP-E4HFS1K:~$ echo "hello srinivas" >> sample.txt
```

400 write permission is denied.

```
srinivas@DESKTOP-E4HFS1K:~$ chmod 400 sample.txt
srinivas@DESKTOP-E4HFS1K:~$ chmod 400 sample_dir
srinivas@DESKTOP-E4HFS1K:~$ ls -l sample.txt
-r----- 1 srinivas srinivas 30 Aug 22 12:21 sample.txt
srinivas@DESKTOP-E4HFS1K:~$ cat sample.txt
hello srinivas
hello again
# Read is allowed
srinivas@DESKTOP-E4HFS1K:~$ echo "new text" >> sample.txt
bash: sample.txt: Permission denied
# Modify is denied
srinivas@DESKTOP-E4HFS1K:~$ echo "another line" > sample.txt
bash: sample.txt: Permission denied
srinivas@DESKTOP-E4HFS1K:~$ ls -l sample.txt
-r----- 1 srinivas srinivas 30 Aug 22 12:21 sample.txt
srinivas@DESKTOP-E4HFS1K:~$ cat sample.txt
hello srinivas
hello again
# File content unchanged
srinivas@DESKTOP-E4HFS1K:~$ █
```

2. 200 (-w-)Try to read and modify the file ____Try to list, enter, and create a file ____

200 (-w-) — Write Only

```
bash: sample.txt: Permission denied
srinivas@DESKTOP-E4HFS1K:~$ chmod 200 sample.txt
srinivas@DESKTOP-E4HFS1K:~$ chmod 200 sample_dir
srinivas@DESKTOP-E4HFS1K:~$ echo "hello srinivas" >> sample.txt
srinivas@DESKTOP-E4HFS1K:~$ cat sample.txt
cat: sample.txt: Permission denied
srinivas@DESKTOP-E4HFS1K:~$ cat sample.txt
cat: sample.txt: Permission denied
srinivas@DESKTOP-E4HFS1K:~$ echo "hello srinivas" >> sample.txt
srinivas@DESKTOP-E4HFS1K:~$ ls -ld sample_dir
d-w----- 1 srinivas srinivas 4096 Aug 22 12:18 sample_dir
```

3. 100 (--x) Try to read and execute the file ____ Try to list and enter the directory ____

100 (--x) — Execute Only

```
srinivas@DESKTOP-E4HFS1K:~$ ls sample_dir
ls: cannot open directory 'sample_dir': Permission denied
srinivas@DESKTOP-E4HFS1K:~$ touch sample_dir/test.txt
touch: cannot touch 'sample_dir/test.txt': Permission denied
srinivas@DESKTOP-E4HFS1K:~$ chmod 100 sample.txt
srinivas@DESKTOP-E4HFS1K:~$ ls -l sample.txt
---x----- 1 srinivas srinivas 30 Aug 22 12:21 sample.txt
srinivas@DESKTOP-E4HFS1K:~$ cat sample.txt
```

4. 500 (r-x) Try to read and modify the file ____ Try to list and enter the directory ____

500 (r-x) — Read + Execute

```
srinivas@DESKTOP-E4HFS1K:~$ cat sample.txt
cat: sample.txt: Permission denied
srinivas@DESKTOP-E4HFS1K:~$ cd ..
srinivas@DESKTOP-E4HFS1K:/home$ cd srinivas/
srinivas@DESKTOP-E4HFS1K:~$ cd sample_dir/
bash: cd: sample_dir/: Permission denied
srinivas@DESKTOP-E4HFS1K:~$ chmod 500 sample.txt
srinivas@DESKTOP-E4HFS1K:~$ cat sample.txt
hello srinivas
hello srinivas
```

5. 600 (rw-) Try to read and modify the file ____ Try to list, enter, and create a file ____

600 (rw-) — Read + Write

```
srinivas@DESKTOP-E4HFS1K:~$ echo "new text" >> sample.txt
bash: sample.txt: Permission denied
srinivas@DESKTOP-E4HFS1K:~$ chmod 500 sample_dir
srinivas@DESKTOP-E4HFS1K:~$ ls sample_dir
inside.txt
srinivas@DESKTOP-E4HFS1K:~$ cd sample_dir
srinivas@DESKTOP-E4HFS1K:~/sample_dir$ cd ..
```

6. **700 (rwx)** Try to read, modify, and execute the file __Try to list, enter, create, and delete file _____

700 (rwx) — Full Access

```
srinivas@DESKTOP-E4HFS1K:~$ touch sample_dir/test.txt
touch: cannot touch 'sample_dir/test.txt': Permission denied
srinivas@DESKTOP-E4HFS1K:~$ chmod 600 sample.txt
srinivas@DESKTOP-E4HFS1K:~$ ls -l sample.txt
-rw----- 1 srinivas srinivas 30 Aug 22 12:21 sample.txt
srinivas@DESKTOP-E4HFS1K:~$ cat sample.txt
```

Questions:

1. **What does read (r) permission allow you to do with a regular file?**

It allows you to **view/read the contents** of the file.

2. **What does read (r) permission allow you to do with a directory?**

It allows you to **read the directory's contents/list of filenames**. It does not by itself allow you to enter or access those files.

3. **What does write (w) permission mean for a regular file?**

It allows you to **modify, overwrite, or write to the file**. It does not automatically allow you to read its contents.

4. **What does write (w) permission mean for a directory?**

It allows you to **add, remove, or rename directory entries**, but normally you also need x permission to actually perform these operations.

5. **What does execute (x) permission mean for a regular file?**

It allows the file to be **executed as a program**, provided its contents and execution environment support that.

6. **Why is execute (x) permission important for accessing a directory?**

Directory x means **search/traverse** permission. It allows you to enter the directory and access entries within it when the relevant permissions are also available.

7. **Which permission combination allows the owner to fully access a regular file?**

700 (rwx) gives the owner read, write, and execute permission.

8. **Which permission combination allows the owner to fully access a directory?**

700 (rwx) gives the owner permission to list, enter/traverse, create, delete, and modify directory entries.

9. **Why do the same r, w, and x permissions behave differently for a regular file and a directory?**

Because a regular file and a directory represent different filesystem objects. For a **file**, r controls its contents, w controls modification of its contents, and x controls execution. For a **directory**, r controls reading its list of entries, w controls changing its entries, and x controls traversal/access through the directory. Thus, the same permission bits have meanings appropriate to the type of filesystem object.